

Raghavendra Pedada

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Professional Summary

CS undergraduate (IoT, Cybersecurity & Blockchain) experienced in full-stack development, AI-powered systems, RAG pipelines, and edge AI. Delivered 4+ end-to-end projects using Python, FastAPI, React, Docker, GitHub Actions, Raspberry Pi, and TFLite. Passionate about scalable software, on-device AI inference, and performance-oriented engineering.

Education

B.Tech, Computer Science & Engineering <i>Specialization: IoT, Cybersecurity & Blockchain</i> Alva's Institute of Engineering and Technology (VTU), Mangaluru	<i>Expected: 2027</i> CGPA: 8.14
Intermediate (Class XII) — Narayana Junior College, Andhra Pradesh	07/2023 78%
Secondary School (Class X) — Sri Chaithanya Techno School, Andhra Pradesh	07/2021 99%

Technical Skills

Languages	Python, Java, C++, C, JavaScript, SQL
Frameworks	FastAPI, React, Vite
Web Tech	HTML5, CSS3, JavaScript, REST APIs
AI/ML & Edge	Prompt Engineering, RAG, LLMs, TensorFlow Lite, MediaPipe, OpenCV
Embedded	Raspberry Pi, ESP32, Arduino, MQTT, Sensors
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Databases	SQLite, MySQL
Libraries	SQLAlchemy, Chart.js, Framer Motion, Axios
Tools	Git, GitHub, Docker, GitHub Actions, Linux, VS Code

Experience

Cybersecurity Micro-Intern <i>Tata Consultancy Services (TCS) — Virtual Programme</i>	<i>2024</i>
- Completed programme covering network security, threat modelling, and vulnerability assessment. - Evaluated IoT device security postures and proposed mitigation strategies for common attack vectors.	

Projects

HashPilot – AI-Powered Computational Benchmark Platform GitHub: GitHub	<i>FastAPI • React • SQLite • Docker • AI</i>
- Developed a full-stack AI-powered computational benchmarking platform to evaluate Proof-of-Work strategies including Sequential, Random, MultiThread, and MultiProcess implementations. - Built a FastAPI backend exposing REST APIs for benchmark execution, analytics, benchmark history, AI-powered strategy recommendations, and system information collection. - Designed an interactive React dashboard with benchmark visualization, analytics, performance charts, winner detection, and responsive user experience. - Integrated SQLite for persistent benchmark storage, Docker for containerization, and GitHub Actions for automated testing and CI/CD workflows.	
Cosmic RAG System GitHub: GitHub	<i>Python • RAG • LLMs</i>
- Built a Retrieval-Augmented Generation pipeline for intelligent document question answering. - Integrated semantic retrieval with Large Language Models to generate context-aware responses. - Designed a modular architecture supporting scalable AI-powered document intelligence applications.	
AI Tool Calling Assistant GitHub: GitHub	<i>Python • APIs</i>
- Developed an AI assistant capable of invoking external tools through structured function calling. - Integrated API-driven workflows to automate multi-step tasks and improve productivity. - Applied modular software engineering principles to build scalable and maintainable AI workflows.	
Edge-AI Smart Mirror	<i>Final Year Project • Raspberry Pi 4B • TFLite • MediaPipe • OpenCV</i>
- Built a real-time, on-device health and wellness monitoring system on Raspberry Pi 4B with zero cloud dependency. - Implemented posture detection, mood analysis, and rPPG-based vitals estimation using MediaPipe and TFLite inference pipelines. - Designed modular OpenCV-based vision architecture enabling concurrent multi-metric tracking at low latency on edge hardware.	
IoT Health Monitoring System	<i>ESP32 • MQTT • Python • Sensors</i>

- Designed an IoT-based health monitoring system using ESP32 microcontrollers and biometric sensors for real-time vital data acquisition.
- Published sensor telemetry over MQTT to a Python backend for processing, logging, and threshold-based alerting.
- Demonstrated end-to-end embedded-to-cloud pipeline bridging hardware and software layers.

Certifications

- **IBM/Coursera** — CI/CD with OpenShift: GitHub Actions & Tekton Pipelines
- **TCS** — Cybersecurity Micro-Internship (Virtual)

Links

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